Toward Sensor-Light Closed-Loop Ultrafiltration Control in Haemodialysis: Retrospective Risk Modelling and Exploratory Action-Conditioned Simulation

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Abstract

Intradialytic hypotension (IDH) is among the most frequent complications of haemodialysis. Clinical closed-loop systems that adjust the ultrafiltration rate (UFR) already exist, but they rely on optical relative-blood-volume sensors that a wearable device cannot easily carry. This project asks whether routinely recorded dialysis-machine data alone can support a predict-then-adjust loop, and what such a loop would cost. Using HEMOBP (CC BY 4.0; 1,072 patients recorded, 919 remaining after cleaning, 160,069 analysable sessions), we predict IDH in the remainder of a session from information available at a decision point T, with patient-level splits and AUPRC as the primary metric. Five models are compared. At the 60-minute decision point the best model reaches AUPRC 0.2015 against a 0.0586 prevalence baseline, but patient-level bootstrap intervals are wide (95% CI 0.13–0.28) and the gap to the compact 1D-CNN+MLP is not established (difference 0.018, 95% CI -0.001 to 0.033), so we do not rank architectures. Raw probabilities are unusable as risks: they over-predict by a factor of four to six and every Brier score is worse than the no-skill value, which Platt scaling on validation repairs (0.052 versus 0.055). In the XGBoost ablation, pre-decision blood pressure carried the largest share of predictive information (removing it costs 44% of AUPRC) while machine-setting features added little independent value under the evaluated feature set. An exploratory action-conditioned simulation, with candidates generated only from the pump setting observed at T, illustrates the trade-off rather than validating a policy: holding that setting implies 7.3% predicted risk with 131 mL residual, while a model-chosen reduction gives 5.9% with 354 mL — paired patient-level bootstraps put that difference at 1.4 points (95% CI 1.1–1.7), but the same model both chose and scored the action, only 65% of candidates have conditional support, and the risk model is conditional rather than identified. The control section is therefore reported as a sensitivity analysis; establishing an intervention effect requires prospective data.

1 Problem Formulation

Haemodialysis replaces kidney function intermittently. Between sessions, patients accumulate fluid; during a four-hour session a machine must remove that fluid at a rate the cardiovascular system can tolerate. When the ultrafiltration rate (UFR) exceeds the capillary refill rate, effective circulating volume falls and intradialytic hypotension (IDH) follows. In our cohort, 11.2% of sessions reach a

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nadir systolic blood pressure (SBP) below 90 mmHg. This is the kind of clinical prediction problem the course frames as a healthcare-AI task (Lecture 1, ~00:40) [17].

Three generations of countermeasures exist. First, reactive control: a proportional–integral–derivative (PID) controller, or nursing judgement, reacts only after blood pressure has already fallen. Second, blood-volume biofeedback: commercial systems such as Hemocontrol (Baxter) and the BVM/BTM-based feedback of Fresenius adjust UFR and dialysate conductivity in closed loop, guided by an optical sensor that measures relative blood volume [4, 15, 16]. Third, machine-learning prediction, which forecasts IDH from clinical variables but has rarely been evaluated as a component of a control loop [1, 14].

For the second generation, some studies report benefit while a recent review concludes that evidence for clinical adoption is still under evaluation [4]. It also depends on an optical relative-blood-volume sensor; on the basis of the design project's own component assumptions that sensor is bulky and costly relative to a wearable envelope, but we did not verify those specifications here, so we treat this as a design motivation rather than an established fact. This motivates the question we study here: *with routinely recorded machine data only — pump settings, dialysate parameters, and the sparse blood-pressure readings nurses already take — can a predict-then-adjust loop be built, and what does it cost?*

We decompose this into three testable sub-questions, each mapped to a claim from an accompanying engineering design project: (i) how early and how accurately can IDH be predicted (claim C1/C2); (ii) is multi-modal sensing necessary, and which modality matters (C3); (iii) what predicted-risk, fluid-removal and session-time trade-offs appear under exploratory action-conditioned simulations (C4/C5) — with a fourth question about whether observational data can support causal claims at all (C6).

The work is deliberately positioned against the course material: irregular EHR time series (Lecture 5, ~00:38), recurrent models (Lecture 6, ~02:14), the Transformer (Lecture 7, ~00:27 and ~27:16), multi-modal fusion (Lecture 8, ~00:11), the reinforcement-learning view of sequential decisions (Lecture 10, ~00:27), interpretability (Lecture 11, ~05:11) and privacy in federated deployments (Lecture 12, ~00:39) [17]. Software 3.0 (generative AI) was used early here, to survey the closed-loop ultrafiltration literature and to test whether the design project's "prediction beats PID" framing was defensible. It was not (Section 2), and correcting that framing changed the research question rather than its wording.

2 Related Work

Closed-loop ultrafiltration. Santoro et al. introduced a multivariable closed-loop controller that adjusts UFR and dialysate conductivity to follow a target blood-volume trajectory, cutting severe hypotensive episodes from 26 to 3 in a crossover study [16]. Ronco et al. found fewer hypotensive episodes and better equilibrated Kt/V under biofeedback [15]. A recent narrative review located 28 clinical studies, of which 23 reported effectiveness and none reported adverse effects, while concluding that evidence for clinical adoption still requires randomised trials [4]. Our work does not compete with these systems; it asks what can be done when their sensor is unavailable.

IDH definitions and epidemiology. Prevalence depends heavily on definition [10], and different definitions carry different mortality associations [5]. Guidelines recommend keeping the ultrafiltration rate below roughly 13 mL/kg/h, alongside ultrafiltration profiling and cooler dialysate [3, 13]. We therefore report three definitions and justify the one we adopt.

Machine learning for IDH. Tree ensembles and recurrent networks dominate the recent literature; a bidirectional LSTM reached 78.5% accuracy and AUC 0.822 on 73,323 sessions [14], and a multi-scale temporal feature model reported 98.1% AUC with XGBoost on a public cohort [1]. A Temporal Fusion Transformer reached AUC 0.953 but at high compute cost. Two gaps motivate our design: prediction is rarely coupled to a control evaluation, and the causal status of the ultrafiltration signal is rarely examined. We address both, and we quantify the fluid-removal price of acting on a prediction. Our modelling choices also follow the course treatment of irregular clinical time series (Lecture 5, ~00:38), recurrent gating (Lecture 6, ~02:14) and attention over short sequences (Lecture 7, ~27:16) [17].

Table 1: Sample flow. Rows 1–5 are the session-level cleaning steps (each row is the number of sessions remaining after that filter); the lower block is the horizon-specific exclusion that turns eligible sessions into modelling samples (`no readings` = 8 sessions present in the weight file but absent from the machine-reading file).

	T=30	T=60	T=90	T=120
sessions reconstructed	197,269	197,269	197,269	197,269
− duration outside 60–480 min	196,457	196,457	196,457	196,457
− start weight outside 20–200 kg	162,563	162,563	162,563	162,563
− interdialytic gain outside −2 to 10 kg	160,673	160,673	160,673	160,673
− mean UFR outside 0–20 mL/kg/h	160,069	160,069	160,069	160,069
eligible sessions	160,069	160,069	160,069	160,069
− session shorter than T+30 min	0	244	567	1,160
− fewer than 3 pre-T readings	12,798	8,468	3,192	1,071
− no post-T readings	30	45	83	155
− already below 90 mmHg before T	3,658	4,260	5,458	6,493
− no machine readings	8	8	8	8
modelling samples	143,575	147,044	150,761	151,182

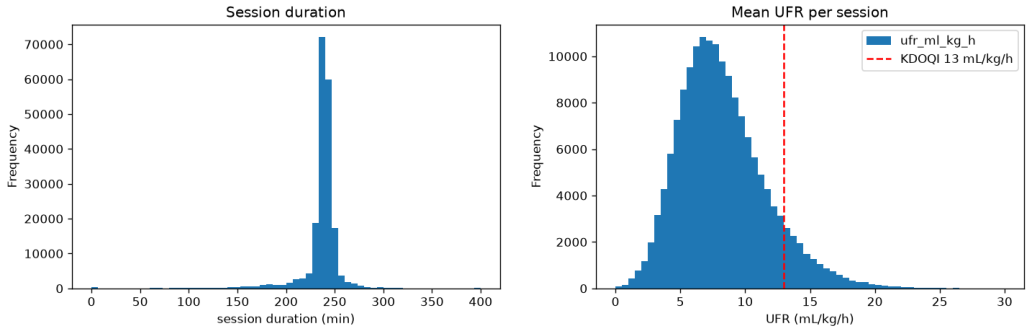


Figure 1: Cohort descriptives. Left: session duration. Right: mean UFR per session against the 13 mL/kg/h guideline threshold.

3 Data

Cohort. HEMOBP [11] contains dialysis-machine recordings linked to hospital demographic data, released under CC BY 4.0: 1,072 patients, 4,366,298 intra-session readings. Sessions are not a calendar day — the unit runs three shifts and the last crosses midnight — so we reconstruct sessions as contiguous runs of readings separated by more than six hours, and assign each to a dialysis day beginning at 08:00. The source cohort thus contains 1,072 patients and 197,269 reconstructed sessions. Eligibility filtering (Table 1) leaves 160,069 sessions from **919 patients** before the horizon-specific exclusions, and it is those 919 patients that every modelling result describes — not the 1,072 recorded. The start-weight filter is the single largest loss and is what removes 153 of them.

Descriptive statistics. Mean session duration is 236 ± 24.5 minutes; mean UFR is 8.26 ± 3.38 mL/kg/h (median 7.8), i.e. mostly below the guideline threshold of 13; median interdialytic weight gain is 2.32 kg. Crucially, blood pressure is sparse: the median within-session interval between distinct readings is 50.7 minutes, and only 32% of intervals are at or below 20 minutes (Figure 1).

Outcome definition. Table 2 reports three candidate definitions. A drop of ≥20 mmHg applies to 84.4% of sessions and is therefore unusable as a target; we adopt nadir SBP <90 mmHg (11.24%) as primary and report nadir <100 mmHg (20.73%) for sensitivity.

Table 2: IDH incidence under three definitions (session level, n=197,269).

Definition	Incidence	Use
nadir SBP $<$ 90 mmHg	11.24%	primary
nadir SBP $<$ 100 mmHg	20.73%	sensitivity
max $-$ nadir $\geq$ 20 mmHg	84.42%	rejected (non-specific)

Sampling design. For each session we form one sample per decision point $T \in \{30, 60, 90, 120\}$ minutes. Inputs use only readings with time $\leq T$; the label is the nadir SBP of readings after T. Sessions already below 90 mmHg before T are excluded, since the prediction would be moot; sessions with fewer than three pre-T readings are excluded. This yields 143,575 (T=30), 147,044 (T=60), 150,761 (T=90) and 151,182 (T=120) samples, with remaining-window IDH rates of 8.0%, 7.7%, 7.1% and 6.5% respectively. As a qualitative example, take one test session at T=60 min: 1.88 L of interdialytic gain, 0.46 L already removed, 1.42 L still to remove, 2.42 h remaining, and a clinician-chosen 0.48 L/h for the remainder. That plan removes 1.16 L and therefore leaves 266 mL unremoved, while SBP drifts from 148 to a nadir of 110 mmHg and no IDH event occurs. Sessions like this are where the label becomes noisy at the margin, and they are also where the fluid trade-off in E5 bites. Cleaning follows pre-registered bounds (duration 60–480 min, weight 20–200 kg, interdialytic gain -2 to 10 kg, UFR 0–20 mL/kg/h): 197,269 sessions reduce to 160,069 eligible.

Limitations of the data. HEMOBP is *in-centre* haemodialysis: four hours, supine, with sparse cuff measurements. It contains no photoplethysmography, electrocardiography, inertial or relative-blood-volume signals, and no symptom or nursing-intervention records. It therefore cannot validate a five-to-fifteen-minute warning window, and it cannot represent the ambulatory wearable scenario described in the accompanying design project. We return to this mismatch in the discussion.

4 Approach

Framing. The primary task is classification: given the state at T, will this session reach a nadir SBP below 90 mmHg afterwards? A risk model conditioned on a proposed plan appears only *inside* the control simulation (Section 4), where it scores candidate plans; it does not identify the counterfactual outcome that would follow if a plan were imposed. Reporting classification as the main result keeps the headline numbers interpretable and avoids over-claiming.

Features and modalities. Fifteen features fall into three groups, which we later ablate: (a) *demographic/static* — age, sex, diabetes, start weight, dry weight, interdialytic weight gain; (b) *pre-T physiology* — first SBP, nadir so far, SBP slope, count of distinct readings; (c) *machine settings* — mean UFR, blood flow, dialysate conductivity and temperature up to T. Sequence models additionally consume up to 96 readings of seven channels (SBP, DBP, UFR, blood flow, conductivity, dialysate temperature, normalised elapsed time) with an explicit padding mask.

Models. Five models are compared. (1) XGBoost [2] on tabular features is the clinical baseline. (2) A multilayer perceptron consumes static features only. (3) A 1D-CNN+MLP fusion mirrors the architecture of the accompanying design project: a convolutional branch over the reading sequence (convolutional feature extraction as developed for medical images in Lectures 2–3, ~00:45 and ~00:39) [17], an MLP branch over static features, concatenated before a shared head. (4) An LSTM [6] reads the masked sequence and takes the last valid hidden state. (5) A Transformer encoder built from scratch reuses the encoder written for the course assignment on clinical sequences [12]: sinusoidal positional encoding, multi-head scaled dot-product attention with key-padding masks, GELU feed-forward blocks and mean pooling over valid positions [18].

Evaluation protocol. Splits are by patient (70/15/15) with a fixed seed, because a patient contributes dozens of sessions and row-wise splitting would leak. AUPRC is the primary metric given class imbalance; AUROC, Brier score and sensitivity at 80% specificity are secondary. Because AUPRC is prevalence-sensitive, we report test prevalence alongside every value.

Training procedure. All four neural models share one training recipe, which reduces tuning-related variation; the models nevertheless consume different inputs (static features only for the MLP, sequence plus static features for the fusion model, sequence-dominant inputs for the LSTM and the Transformer), so E1 compares complete modelling pipelines rather than isolating architecture alone. Inputs are standardised with training-split statistics only: each sequence channel is centred and scaled over valid (non-pad) training entries and re-zeroed at padded positions, and the fifteen static features are standardised with training-split mean and standard deviation. Training minimises binary cross-entropy with `pos_weight` set to the training-split negative-to-positive ratio (17.1:1 at T=60); optimisation is Adam with learning rate 10^{-3} and batch size 256, for at most ten epochs, with early stopping of patience two epochs on validation AUPRC and restoration of the best-validation checkpoint before test evaluation. Dropout is 0.2 in the fusion and head layers and 0.1 inside the Transformer encoder, which uses d=64, four heads and two layers. Python, NumPy and PyTorch seeds are all fixed to 42. XGBoost uses the histogram tree method, 600 trees of depth 6, learning rate 0.05, 0.8 row and column subsampling, and `scale_pos_weight` equal to the same ratio. These defaults are revisited in E7.

Control policies (in silico). The control analysis rests on an *action-conditioned risk model*, not on an identified counterfactual model. It adds two features to the tabular set — a planned remaining UFR and the remaining minutes — and is fitted on pairs of (state at T, the UFR clinicians actually went on to deliver). It therefore estimates $P(\text{IDH} \mid \text{state}, \text{plan})$ on the distribution of plans clinicians chose; it does not identify what would happen under a plan that was rarely or never chosen. Arms are generated only from what a controller knows at T: the *pump setting observed at* T (the last non-zero setting at or before T) is the anchor, and candidates are $\{1.0, 0.9, 0.8, 0.7, 0.6\}\times$ that setting. Each candidate is then entered into the model as the planned remaining UFR, which is a simplification: it assumes the plan would be held, whereas a real device would react continuously. The arms are *hold* (keep the setting), *reactive* (cut 30% if SBP has fallen $\geq$15 mmHg or nadir is below 100), and *predictive* (minimise predicted risk subject to a residual-fluid tolerance). The trajectory clinicians followed is reported only as a descriptive reference, because it was not a choice available at T. All quoted risks are Platt-recalibrated with parameters fitted on the validation split. This section is a sensitivity analysis of the trade-off; it is not a validation of a control policy.

Causal plausibility diagnostics. Two probes examine the internal consistency of the ultrafiltration signal and its exposure to confounding; they do not identify a causal effect. First, a perturbation sweep: we vary the planned remaining UFR multiplier from 0.3 to 1.3 and record mean predicted risk, plus the crude observed association between actual UFR and IDH. Second, an intervention-style ablation on the Transformer: zeroing the attention output or the feed-forward output of each encoder block (the intervention tooling follows 19) and measuring the change on the logit scale (not the probability scale, which saturates). Figure 6 shows the perturbation and association results.

Software 3.0. Generative AI assistance was used throughout, under the course honour code, and is disclosed here. It was most valuable in three places: refactoring the session builder after pandas-groupby iteration failed on 160k sessions (a vectorised rewrite cut runtime from >20 min to 19 min for four horizons); diagnosing a misleading ablation result in an earlier assignment that turned out to be sigmoid saturation rather than a modelling failure; and localising a device-placement bug in a diagnostic model. It was least reliable on domain judgement — choosing the IDH definition, deciding what the control comparison may legitimately claim, and recognising that the biofeedback literature makes "predictive versus PID" the wrong framing. All numerical results in this report were produced by scripts in the accompanying repository and can be regenerated.

5 Experiments

E1: five-model comparison. Table 3 reports the 60-minute decision point on 22,407 test samples (the label covers the remainder of the session, not the following 60 minutes) (prevalence 5.86%). The from-scratch Transformer has the highest point estimate, but patient-level bootstrap intervals are wide and overlap: it exceeds the 1D-CNN+MLP by 0.018 with a 95% interval of -0.001 to 0.033 (Figure 2). We therefore *do not* claim an architecture ranking; what the comparison supports is that a compact fused model is competitive at a quarter of the parameters. Two cautions follow. The validation AUPRC values used for checkpoint selection (up to 0.32) sit above the test values

Table 3: E1: five models at T=60 min (test n=22,407, prevalence 5.86%). Intervals are patient-level bootstrap (B=1,000). Brier scores are raw and all exceed the no-skill 0.0552 (E8).

Model	AUPRC	95% CI	AUROC	Sens@80%spec	Brier	Params
XGBoost	0.1675	0.118–0.221	0.733	0.548	0.116	—
MLP	0.1721	0.107–0.257	0.758	0.562	0.150	10,369
1D-CNN+MLP	0.1837	0.117–0.268	0.768	0.573	0.136	24,961
LSTM	0.1804	0.119–0.247	0.761	0.569	0.175	32,129
Transformer	0.2015	0.131–0.282	0.774	0.587	0.164	107,265

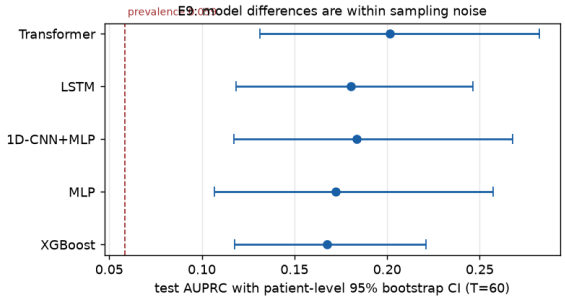


Figure 2: E9: test AUPRC with patient-level 95% bootstrap intervals. The intervals overlap across all five models, so the architecture ordering in Table 3 is not established.

reported here; validation carries 24,665 samples at 6.29% prevalence against 22,407 at 5.86% on test, so a prevalence difference accounts for part of the gap, and selecting the best of ten checkpoints on validation accounts for another part. We report both numbers rather than attribute the gap to a single cause. And every Brier score exceeds the no-skill value $p(1-p) = 0.0552$, so they cannot be read as calibration quality; E8 diagnoses why and repairs it.

E2: compute budget. Table 4 shows why the fusion model is the reasonable deployment choice: 0.271 ms per sample on CPU and 3.31 ms for a batch of 256, against 1.487 ms and 147.2 ms for the Transformer — 44× faster at batch scale, while remaining competitive under the default comparison settings (E1/E9). Latency was measured on a laptop (RTX 3070 Ti, CPU fallback) and is indicative, not an embedded-MCU benchmark. LSTM FLOPs are under-counted by the profiler because recurrent terms are not attributed; we do not compare on that column.

E3: prediction horizon. Table 5 shows a non-monotone pattern: quality peaks at T=90 min for the fusion model (AUPRC 0.2395, AUROC 0.807). Two effects trade off — later decision points have more observed history, but their remaining windows contain fewer events, so prevalence falls from 6.21% to 4.75%. Cross-T comparisons therefore mix window and prevalence effects and must not be read as "later is better". The practically important point is the floor: with blood pressure sampled every 50.7 minutes at the median, prediction horizons below roughly one sampling interval cannot be validated on this data. The five-to-fifteen-minute warning requirement specified in the accompanying design project therefore cannot be validated at this sampling frequency. Continuous physiological sensing, including photoplethysmography, remains a hypothesis for future investigation; its incremental predictive value would have to be established on data that actually contain such signals.

E4: modality ablation. Table 6 answers the sensor question. Pre-T blood-pressure history alone nearly matches the full feature set (0.1592 versus 0.1675), and removing it costs 44% of AUPRC — the largest single loss. Machine settings alone are weak (0.0751), and removing them slightly *improves* the model (0.1740), indicating little independent signal at T=60; this is consistent with clinicians already reacting to blood pressure, so the UF setting partly encodes that reaction. Static demographics contribute a small but non-zero amount. Within this data, then, physiological sensing carries the signal and additional machine-side channels do not. We stop short of the stronger claim that adding a high-frequency sensor such as photoplethysmography would improve prediction: the

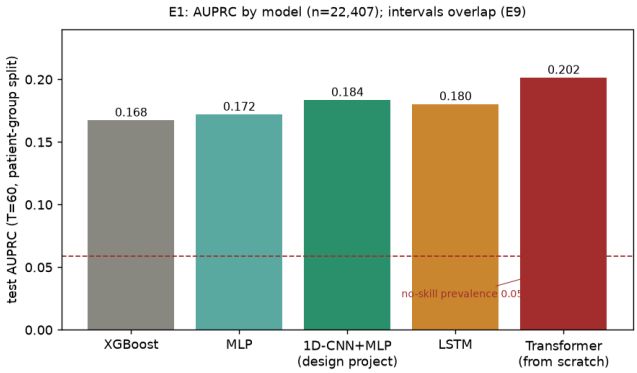


Figure 3: E1: AUPRC by model against the no-skill prevalence line.

Table 4: E2: accuracy versus cost (T=60). Latency on CPU; FLOPs per sample.

Model	Params	FLOPs	Lat. bs1	Lat. bs256
XGBoost	600 trees	—	0.003 ms	—
1D-CNN+MLP	24,961	1.22 M	0.271 ms	3.31 ms
LSTM	32,129	0.06 M*	0.544 ms	9.85 ms
Transformer	107,265	23.69 M	1.487 ms	147.2 ms

current data contain no such signal, so that remains a hypothesis for the next study rather than a result here. Caveat: this ablation uses XGBoost; sequence models might extract more from the dense machine stream, which we leave as follow-up.

E5: exploratory action-conditioned risk simulation. This section is a sensitivity analysis, not a test of a control policy, and we label it accordingly. Two earlier versions were withdrawn: the first anchored candidates on the UFR actually delivered *after* T, which is unavailable at decision time; the second anchored them on the mean rate delivered so far, which is not the rate the machine is running. The current anchor is the **pump setting observed at** T (the last non-zero setting at or before T; mean 0.70 against 0.62 for the historical mean, correlation 0.80), with candidates $\{1.0, 0.9, 0.8, 0.7, 0.6\}\times$ that setting. Arms are *hold*, *reactive* (cut 30% if SBP has fallen $\geq$15 mmHg or nadir is below 100), and *predictive* (minimise predicted risk subject to a residual-fluid tolerance). The plan clinicians followed is a descriptive reference only. Risks are Platt-recalibrated, fitted on validation.

Table 7 reports the arms at T=60 min with patient-level bootstrap intervals. The ordering is hold (7.3%) $>$ reactive (7.2%) $>$ predictive (5.9%), and it costs fluid to move down that ordering: 131 mL residual for hold, 239 mL for reactive, 354 mL for predictive. "Extra time" is the additional minutes at the chosen rate needed to remove the fluid the arm leaves behind, measured from the decision point: $\max(0,\, R - uh)/u \times 60$, where R is the fluid still to remove at T (litres), u the chosen UFR (L/h) and h the remaining hours. The reference duration is the observed remaining session time, not a fixed four hours. Marginal intervals overlap, which on its own would be the wrong test; paired patient-level bootstraps on the risk *differences* put hold above predictive by 1.41 points (95% CI 1.12–1.74) and reactive above predictive by 1.30 (1.06–1.60), while hold and reactive are indistinguishable (0.11, -0.06 to 0.25). So the model does order the arms; what it cannot do is establish that the ordering would survive in patients, because the same model both chose and scored the action, and because 65% conditional support leaves a third of the candidates outside comparable practice.

Three further limits. (i) *Conditional* overlap is weaker than marginal overlap: 88% of candidates lie inside the overall observed range of post-T plans, but only 65% lie inside the 10–90 percentile of plans used for patients in the same pre-T nadir and interdialytic-gain stratum. That is a stratification on two pre-decision covariates, not a demonstration of causal overlap: most of the apparent marginal support is between-patient heterogeneity. (ii) The same model both selects and scores the action, which biases the selected arm optimistically; a second, independently seeded model (depth 3, no

Table 5: E3: horizon study. 1D-CNN+MLP AUPRC/AUROC by decision point.

T (min)	30	60	90	120
Test prevalence	6.21%	5.86%	5.20%	4.75%
XGBoost AUPRC	—	0.1675	0.1965	—
1D-CNN+MLP AUPRC	0.1963	0.1837	**0.2395**	0.2266
1D-CNN+MLP AUROC	0.757	0.768	0.807	**0.824**

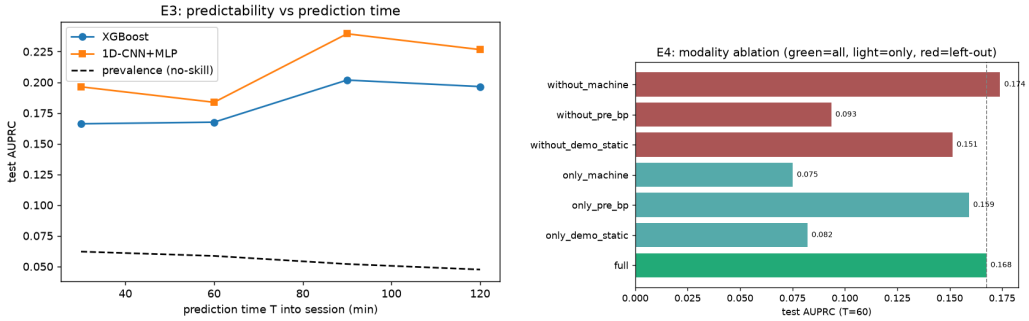


Figure 4: Left: E3 horizon curve against the prevalence (no-skill) line. Right: E4 modality ablation.

class weight) reproduces the ordering but shifts every level up by 0.6–1.3 points, with per-sample correlations of 0.85–0.87. (iii) Calibration is established on the factual distribution; it says nothing about calibration under counterfactual treatment, so the calibrated numbers remain conditional-risk estimates.

E6: sensitivity and causal-plausibility diagnostics. Figure 6 reports both probes. The perturbation sweep is monotone and in the expected direction: raising the planned remaining UFR multiplier from 0.3 to 1.3 raises mean predicted risk from 0.183 to 0.265 (gradient $+0.089$). The crude association agrees: binned by actual post-decision UFR, observed IDH rises from 3.1% to 11.8%. This is necessary for the control simulation to make sense — but it is not sufficient to establish causality, because confounding by indication predicts the same pattern: patients who are more fluid-overloaded, and therefore more fragile, receive higher ultrafiltration and are also more likely to become hypotensive. The monotone response is an internal consistency check for the model-based simulation — it shows the model reacts to UFR in the expected direction — but it does not validate predictions for actions that were not taken, nor does it establish a causal effect; its magnitude cannot be decomposed into causal and confounded parts from this data.

A second probe is more surprising. Zeroing the attention output of each Transformer encoder block changes the logit by about 10^{-8} — effectively nothing — whereas zeroing the feed-forward output changes it by 0.042 and 0.067 at layers 0 and 1. The predictive signal therefore flows through the feed-forward path, not through attention, over a sequence of roughly 22 readings with sparse blood pressure. Two consequences follow: attention-heavy architectures are hard to justify here (reinforcing E1/E2), and attention maps must not be presented as clinical explanations — the same observation-versus-intervention lesson from our earlier coursework, and a caution anticipated by the attention variants discussed in Lecture 9 (~00:53) [17].

E7: hyperparameter sensitivity. Hyperparameters are selected on the validation split. Test scores are reported for every swept setting for transparency, but they were never used to choose one; in fact the two splits rank the settings differently. For XGBoost, tree depth dominates validation AUPRC: depth 3 reaches 0.32 and depth 9 falls to 0.18, with 200 trees preferred over 600 or 1200 (Figure 7). The validation-selected configuration (n=200, depth 3) scores 0.186 on test, against 0.168 for the default (n=600, depth 6). For the 1D-CNN+MLP, validation selects learning rate 10^{-3} (0.345), whose test score is 0.188; the setting with the best test score, 3×10^{-4} (0.194), would not have been selected, and 3×10^{-3} falls to 0.168. Disabling the positive-class weight costs about five relative percent (0.188 to 0.178).

Table 6: E4: modality ablation (XGBoost AUPRC, T=60). "Only" uses one group; "w/o" drops one group.

Feature set	AUPRC
All 15 features	0.1675
Only pre-T physiology	0.1592
Only demographics/static	0.0821
Only machine settings	0.0751
Without pre-T physiology	0.0935
Without demographics/static	0.1513
Without machine settings	0.1740

Table 7: E5: exploratory action-conditioned risk simulation at T=60 min (tolerance 0.5 L). Intervals are patient-level bootstrap (B=500). Arms are model outputs under a conditional-risk model. Marginal intervals overlap, but paired patient-level differences do exclude zero (see text); neither establishes a treatment effect.

Arm	Predicted risk (95% CI)	Residual fluid (95% CI)	Extra time
Observed plan (reference)	5.8% (4.9–7.0)	699 mL (599–830)	120 min
Hold setting at T	7.3% (6.2–8.6)	131 mL (89–190)	13 min
Reactive rule	7.2% (6.1–8.5)	239 mL (184–312)	28 min
Predictive	5.9% (5.0–7.0)	354 mL (288–453)	50 min

One caveat matters more than the numbers. Validation and test disagree about the *ranking* of these configurations — the configuration with the best test score (n=600, depth 3, 0.190) ranks below the best validation score (n=200, depth 3) — which is the same instability that inflates validation AUPRC in E1. At this sample size, hyperparameter selection is not stable enough to support a claim that tuning improves the model; we therefore report E7 as a sensitivity analysis rather than a tuning result, and E1 keeps identical defaults across models so that the comparison holds the pipeline constant.

E8: are the probabilities usable as risks? No, not before recalibration. Table 3 already shows that every Brier score exceeds the no-skill value $p(1-p) = 0.0552$, and Figure 8 shows why: mean predicted probabilities are 0.24–0.35 against an observed 0.0586, an over-prediction of four to six times, with calibration slopes of 0.56–0.82 rather than 1. The cause is the training recipe, not the architecture: `scale_pos_weight` (and `pos_weight`) reweights the loss by 17:1, which shifts the effective prior, and no step maps the resulting scores back onto the observed event rate. Fitting standard two-parameter Platt scaling — a slope and an intercept on the logit, both estimated on the validation split — and evaluating it *on the untouched test split* repairs the probabilities without touching the ranking: Brier falls to 0.0515–0.0525 for all five models, below the no-skill 0.0552, while AUPRC is unchanged by construction. The improvement is real but modest — a Brier skill score of only 4.9–6.7% relative to the no-skill forecast — so the correct reading is that the probabilities carry some information, not that they are accurate enough to drive a clinical decision. Nor does calibration transfer to counterfactual actions: the map is fitted on the distribution of treatments clinicians chose, so nothing here licenses calibrated risk under a treatment that was rarely or never given. Two consequences follow. First, "best Brier among five" in E1 does not mean well calibrated — it means least bad before repair. Second, every risk level quoted in E5 is recalibrated; raw scores would have overstated risk by roughly a factor of four, and a control policy optimising raw scores would be optimising a distorted objective.

E9: is the model ordering real? No. Bootstrap resampling by patient (B=1,000) gives 95% intervals of 0.118–0.221 (XGBoost), 0.117–0.268 (1D-CNN+MLP) and 0.131–0.282 (Transformer); the paired difference between the Transformer and the fusion model is 0.018 with an interval of -0.001 to 0.033, positive in 96.6% of resamples. A 96.6% one-sided figure is suggestive but the interval crosses zero, and after the hyperparameter sweep in E7 the gap narrows to 0.009. We therefore report the comparison as "competitive at a quarter of the parameters" rather than "better".

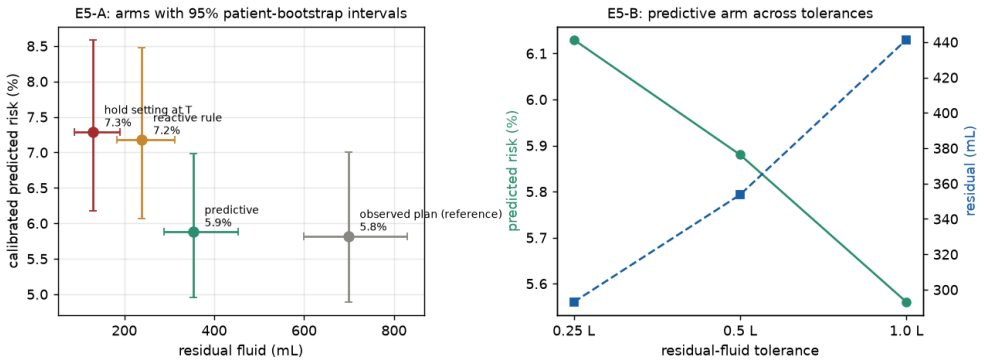


Figure 5: E5: the IDH-versus-fluid trade-off (left) and its sensitivity to the residual-fluid tolerance (right).

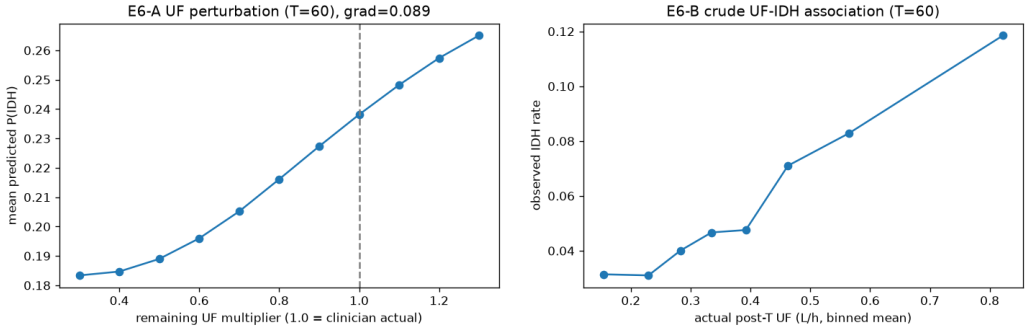


Figure 6: E6: left, predicted risk under UFR perturbation; right, crude observed UFR–IDH association. Both are consistent with a causal effect and with confounding.

Any product decision that hinges on the ranking would need either far more patients or an external cohort.

Software 3.0 in the experimental workflow. Two of the seven experiments exist because of an AI-assisted check rather than a planned one: the probability-versus-logit diagnosis that produced the whole-test-set sanity check, and the suggestion to test attention against feed-forward ablation once the attention heatmaps looked uniform. In the other direction, the first session builder proposed for this project died on 160k sessions; the vectorised rewrite was suggested, but confirming that no patient crossed a split still had to be done by hand. The recurring pattern was that assistance shortened the distance from a symptom to a candidate cause, and contributed nothing to deciding which claim the evidence licenses.

6 Discussion and Conclusion

What we can now say. Three questions were asked at the outset. First, how early and how accurately: with routine machine data and sparse cuff pressures, IDH is predictable at a 60-minute horizon with AUPRC 0.2015 against 0.0586 prevalence (AUROC 0.774) — but with an interval of 0.13–0.28, and no established advantage over a compact fused model at a quarter of the parameters. Second, whether multi-modal sensing is necessary: in the XGBoost ablation at T=60, and within the evaluated feature set, pre-decision blood pressure carried the largest share of predictive information while machine settings added little independent value. That is a statement about this model, these features and this horizon — not about sensing requirements in general. Third, what prediction buys and costs: within the simulation, moving from holding the observed rate to a model-chosen reduction lowers predicted risk from 7.3% to 5.9% while leaving 223 mL more fluid behind, and the intervals overlap — the model produces a consistent predicted-risk–fluid trade-off within this

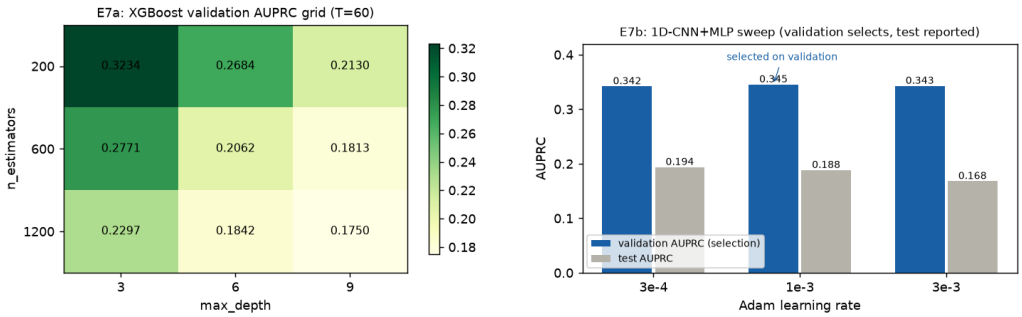


Figure 7: E7: hyperparameter sensitivity. Left: XGBoost validation AUPRC over tree depth and ensemble size (the selection split). Right: 1D-CNN+MLP learning-rate sweep, with validation and test AUPRC shown side by side — selection used validation only; the two splits do not rank the settings identically.

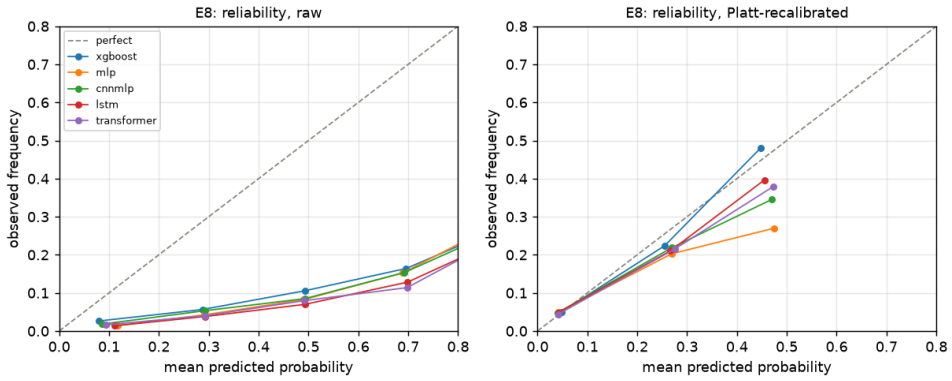


Figure 8: E8: reliability diagrams before (left) and after (right) Platt recalibration. Raw predictions fall below the diagonal, which denotes overprediction; two-parameter Platt scaling improves their alignment with the diagonal on the held-out test set, though it does not make them coincide. Because the raw slopes were 0.56–0.82 rather than 1, the slope term is doing real work: an intercept-only correction would not have fixed the compression.

simulation, but neither its direction nor its magnitude is identified as a treatment effect, because the model is conditional rather than counterfactual.

What the operating point would mean clinically. AUPRC does not by itself establish clinical value. At 80% specificity the Transformer reaches 0.587 sensitivity; at the test prevalence of 5.86% this implies a positive predictive value of about 15%, i.e. roughly 3.4 true events and 18.8 false alarms per 100 sessions. Whether that is usable depends on what an alarm triggers — a nurse review, a patient prompt, an automatic reduction — and on the relative cost of a missed hypotensive episode versus a repeated false alarm. Neither cost is observable in this dataset: there are no symptom records, no intervention logs and no workflow data. The threshold, the alarm burden per session or per hour, and the action that follows therefore remain design decisions to be settled in a prospective study, not quantities this analysis can supply.

Boundaries we will not overstep. Every control number here is in silico under a model trained on clinicians' own decisions; it is not clinical validation. The dose–response gradient is equally compatible with confounding by indication, so the simulated differences are neither estimates nor bounds of a treatment effect: the direction and magnitude of any real clinical benefit remain unidentified in these observational data. The attention-ablation result means explanation should not be read off attention weights. And the cohort is in-centre haemodialysis with a cuff measurement every 50.7 minutes: it cannot speak to the ambulatory wearable scenario, which lacks exactly the

Table 8: What this report does and does not establish. "Established" here means internal validation on one cohort, nothing more; the remaining layers are listed so that the report's scope is explicit.

Layer	What this report provides	Status
Data and algorithm	Patient-level splits, five models, calibration, bootstrap intervals, one cohort	established
Clinical predictive value	One operating point: 0.587 sensitivity at 80% specificity, PPV $\approx 15\%$, 18.8 false alarms per 100 sessions	quantified
Control algorithm	Exploratory action-conditioned simulation; paired risk differences; 65% conditional support; action chosen and scored by the same model	exploratory
Closed loop and device	No pump response, no fault handling, no real-time behaviour	none
Clinical translation	No prospective evaluation, no workflow study	none
Product and business	Outside the scope of a technical report	out of scope

high-frequency modalities (photoplethysmography, inertial, pressure waveforms) whose value this dataset cannot measure — the ablation says blood pressure history matters, not that a new sensor would help. Scaling from 4-hour supine sessions to 6–8-hour ambulatory therapy, and from sparse to continuous sensing, is not one limitation among several but the largest technical assumption in the programme: every number here is conditional on a setting the target product will not reproduce.

Implications for the design project. The results strengthen three claims and weaken one. Strengthened: physiological information is where the predictive signal sits within this dataset, which supports studying how it should be captured rather than settling a sensor budget (E4) — the incremental gain, cost and practicality of continuous monitoring or photoplethysmography need their own data and experiment; a compact convolutional-plus-MLP model is a credible deployment shape, since it is competitive under the default comparison while being far cheaper (E1/E2); and a clinician-set safety envelope with a hard physical override is not bureaucratic decoration but a necessary condition, precisely because the causal strength of ultrafiltration cannot be established retrospectively (E6). Weakened: the design project's framing of its contribution as "prediction beats PID" is historically inaccurate — 28 clinical studies of closed-loop ultrafiltration biofeedback predate it [4]. The more defensible framing is narrower: *testing whether routine machine data and inference can reduce reliance on an optical relative-blood-volume sensor*. This study provides grounds to research that question; it does not show that routine data can replace the sensor, nor that adding photoplethysmography would improve prediction. The five-to-fifteen-minute warning specification also needs restating, since it is not verifiable on routine data and depends on the continuous sensing the device is meant to provide.

What the project taught me. The value of a healthcare-AI contribution sits less in the highest AUPRC than in knowing which claim the data can carry. The most useful result here is a negative one — that the control policy cannot be validated retrospectively — and it took a controlled model comparison, a modality ablation and an intervention probe to establish it rather than assert it.

Future work and intended control function. The long-term objective is a closed-loop ultrafiltration (UF) control function for home or ambulatory haemodialysis. At each control update the intended system will acquire time-aligned, quality-checked physiological signals and machine state; estimate the risk of a clinically meaningful intradialytic hypotension (IDH) event over a validated warning horizon; and select a feasible UF-rate adjustment that balances predicted risk against prescribed fluid removal and acceptable treatment duration. The controller is intended to operate inside clinician-defined treatment limits with independently enforced safety constraints, bounded rate changes, verification of pump response, event logging, and a predefined fallback when signals are missing, stale, inconsistent or unavailable. The 5–15-minute warning requirement remains a target for future validation, not a capability established by this dataset, and the incremental value of continuous blood-pressure monitoring, photoplethysmography (PPG) or other sensors must be evaluated rather than assumed.

Table 9: Design inputs that must be fixed before controller development can start. None is settled by this report, and none should be back-filled from the scores the model happens to reach.

Design input	Why it is still open	Who decides
Intended use: population, modality, setting	In-centre supervised versus home ambulatory changes the sensor set, session length and regulatory route	clinical + product
Control authority: advise, prompt or actuate	Decides whether the model is decision support or a controller, and with it the evidence burden	clinical + regulatory
Hard safety constraints	Rate limits, fluid target, timeouts and signal-loss behaviour must come from hazard analysis, not from model behaviour	control engineering + clinical
Acceptable performance	Warning horizon, missed events, false alarms and added time must come from clinical need, not from AUPRC	clinical + risk analysis

What the present work contributes. This study is a technical foundation for that programme. It establishes a reproducible retrospective pipeline, compares five modelling approaches under patient-level evaluation, identifies substantial probability distortion caused by class-weighted training, and shows why recalibration and uncertainty analysis are not optional. The modality ablation identifies pre-decision blood-pressure history as a major source of predictive information within the evaluated XGBoost feature set. The exploratory action-conditioned simulation makes the trade-off between model-predicted risk, residual fluid and treatment time explicit, while exposing the limits created by incomplete treatment overlap and by selecting and scoring an action with the same model. These findings narrow the questions for the next stage — which signals are needed, what evidence must precede using a risk estimate to guide treatment, and how safety constraints should govern automatic actions — and provide a baseline against which later models and controllers can be measured. They do not establish that the simulated risk differences are treatment effects.

Validation priorities. First, predictive and clinical validation should be strengthened. The full preprocessing and evaluation pipeline should be reproduced on an independent cohort or a temporally separated dataset, keeping patients separate through model selection and evaluation, and should report discrimination, calibration, uncertainty and performance across clinically relevant subgroups. The operational endpoint used here — post-decision nadir SBP below 90 mmHg — should be compared with richer outcomes where available, including symptomatic episodes and events requiring intervention. Candidate operating thresholds should be reported with sensitivity, positive predictive value, false-alert burden and lead time; without those numbers it is impossible to say whether the model adds actionable information beyond routine monitoring rather than merely improving retrospective ranking.

Second, the sensing and warning-horizon requirements should be tested directly. A dataset with sufficiently frequent, time-aligned physiological measurements is needed to evaluate detection within the intended 5–15-minute window. Candidate signals, including continuous blood pressure and PPG where feasible, should be compared with the machine-data baseline using prespecified ablation and incremental-value analyses, while characterising missingness, signal quality, motion artefact and the differences between in-centre and home treatment. The present ablation supports further investigation of physiological information; it does not establish that adding any particular sensor will improve prediction.

Third, action-effect estimation and control-policy development should be separated. The current model estimates risk conditional on observed state and a proposed plan; it does not identify the counterfactual outcome of imposing that plan. Future work should define the treatment-effect estimand explicitly, assess conditional overlap using a broader set of pre-decision covariates, and quantify sensitivity to unmeasured confounding; where data and assumptions allow, causal or target-trial-style analyses could estimate the effect of alternative UF decisions. In parallel, the controller should be evaluated in an independently validated dynamic simulation, with an objective that balances IDH risk, residual fluid and treatment time under hard clinical constraints, rather than selecting the lowest score from the model used to evaluate it. Sensitivity analyses should ask whether a policy remains feasible and stable across plausible patient states, model uncertainty and treatment scenarios.

Staged engineering and clinical verification. Implementation should proceed in stages. The controller should first undergo software-in-the-loop testing and, where feasible, hardware-in-the-loop evaluation on representative pump hardware, covering rate limits, total fluid-removal constraints, delayed or missing measurements, model failure, actuator mismatch, communication loss and recovery from abnormal states. Computational latency should be measured on the intended hardware rather than inferred from laptop benchmarks. A shadow-mode evaluation should then generate recommendations without changing treatment, so that disagreements, unsuitable recommendations and potential safety issues can be characterised. The engineering evidence for those stages is defined outside this report: software lifecycle to IEC 62304 [7], risk management to ISO 14971 [9], and basic safety of haemodialysis equipment to IEC 60601-2-16 [8]; in Singapore the intended-use classification and clinical evaluation of software as a medical device would follow HSA guidance. We cite these as the standards that will govern the work, not as evidence that this prototype meets them. Home deployment also raises data-governance questions — on-device inference, or federated and privacy-preserving training (Lecture 12, ~00:39) — which fall outside this study but condition the design [17]. Only after technical, safety, ethical and regulatory review should a prospective supervised evaluation assess feasibility and safety, followed by controlled studies of whether automatic adjustment improves outcomes without compromising fluid removal or treatment completion.

Intended endpoint. The programme aims at a verified control function that repeatedly senses patient and machine state, estimates near-term IDH risk, proposes and executes bounded UF-rate adjustments, checks the system response, and returns to a safe state when reliable control cannot be maintained. This study is a first step toward that endpoint: it establishes an empirical baseline, identifies limitations that would otherwise be carried into control development, and defines the questions that must be answered before the function can be considered effective or safe. Its contribution is not a claim of treatment benefit but a structured technical basis for building and independently evaluating the next controller.

Conclusion. What this study delivers is narrower than a closed-loop device: an offline risk-prediction pipeline and an exploratory action-conditioned sensitivity analysis, both reproducible on routine data, plus a quantified account of the predicted-risk–fluid trade-off and of why the data cannot settle the causal question. Sensor-light closed-loop control remains a hypothesis for development — one with a clear evidence ladder (external validation, alarm burden, target-device observation, clinician-in-the-loop decisions, and only then automated control), not a demonstrated capability.

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